Artificial intelligence in infectious diseases and clinical microbiology: a scoping review and evidence map of diagnostic, therapeutic and educational applications.

AUTHOR LIST

Smit, MD; Smit R; Ismadi, F; Bakierzynska, M; Hunter, M; Ferreira, J; O'Riordan, R.

SCAN ME

BACKGROUND

Artificial intelligence (AI) is progressively recognised for its emerging role in diagnosing, treating and monitoring health conditions[1,2]. AI has been adopted in infectious diseases (ID) and microbiology to contend with mounting data volumes and complexity, which can outstrip human and classical analytical capacity[3-5].

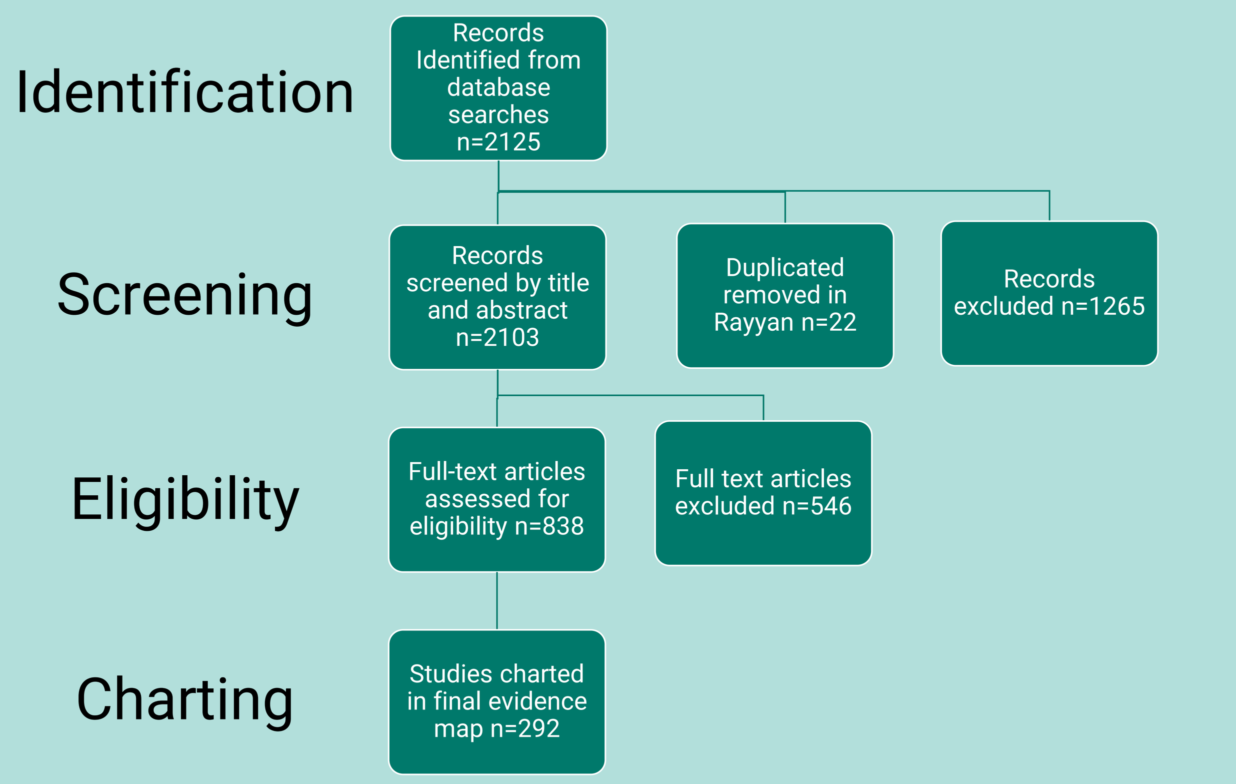
OBJECTIVE

This scoping review aimed to map how artificial intelligence is being applied in infectious diseases and clinical microbiology. Three pre-specified axes were examined: publication trends over time, AI application type and method, and the infection, condition, or microbiology focus of each study. The review sought to identify where evidence is concentrated, where important gaps remain, and how AI use is distributed across diagnostics, therapeutic and prognostic decision-support, and education.

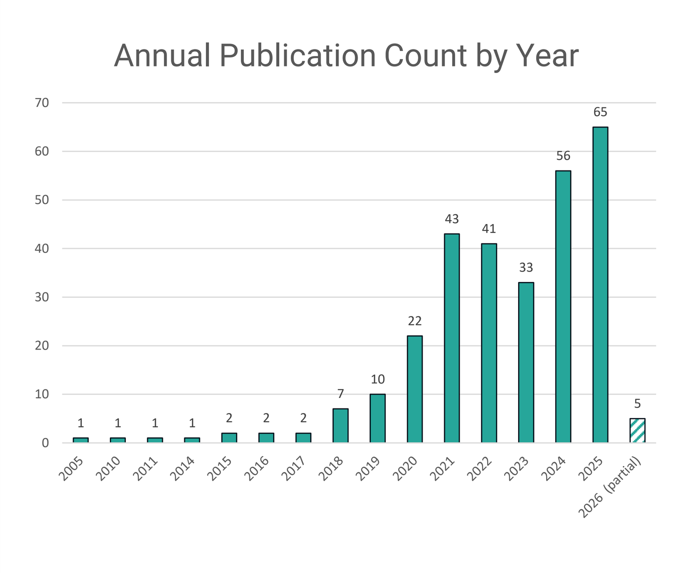
METHODS

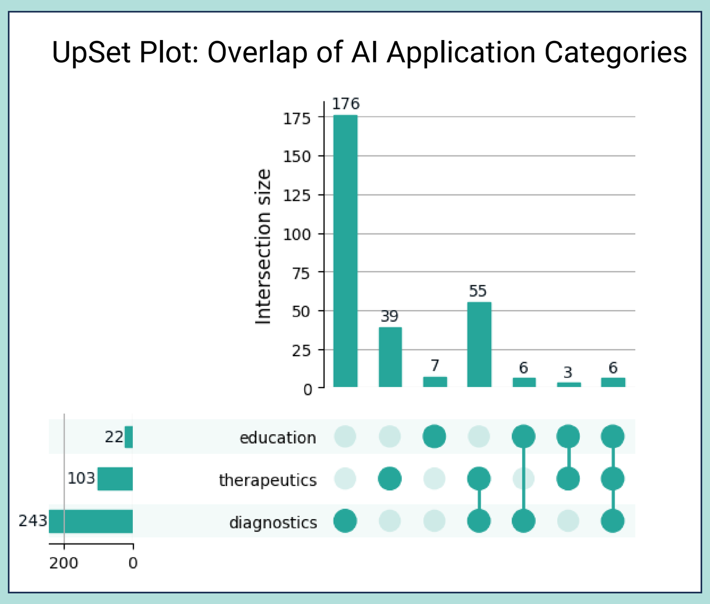
Following JBI methodology for scoping reviews[6], MEDLINE, Embase, and Web of Science were searched. Records were deduplicated and screened in Rayyan[7]. Full texts were assessed against eligibility criteria, and included studies were extracted and labelled across three pre-specified evidence-mapping axes: temporal trends, AI application type and method, and infection or microbiology focus.

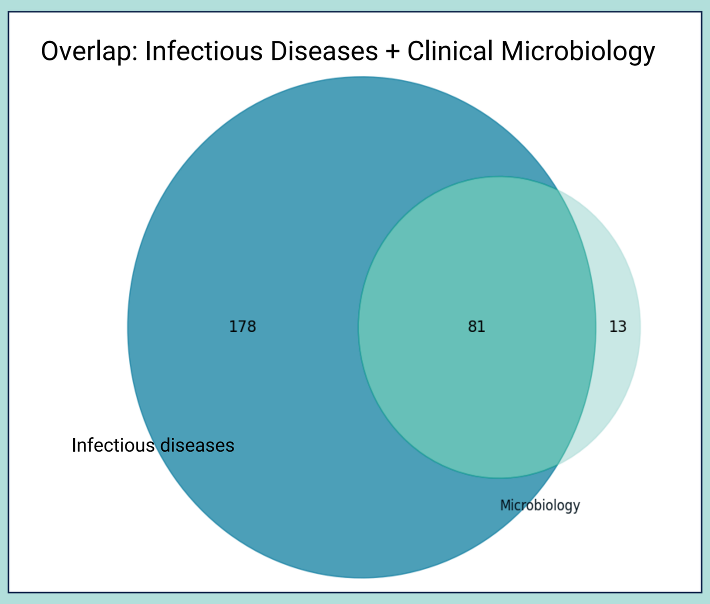
PRISMA-ScR FLOW DIAGRAM

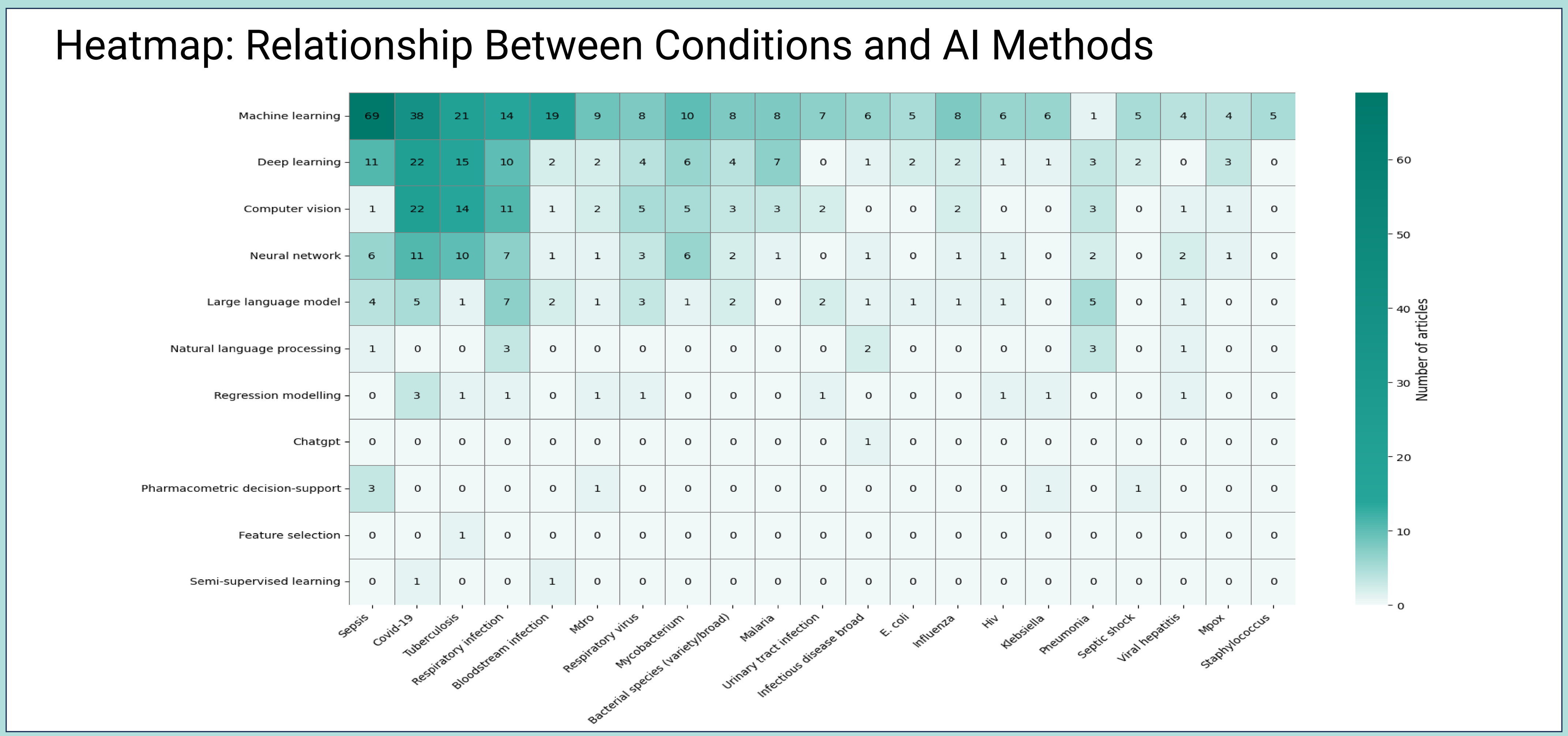


RESULTS









Publications increased markedly over time: 265 studies (90.8%) were published from 2020 onwards. Most studies related to clinical infectious disease practice (n=259, 88.7%), while 90 (30.8%) included a microbiology or laboratory component. Diagnostic applications predominated (n= 243, 83.2%), followed by therapeutic, prognostic, or clinical decision-support applications (n=103, 35.3%); Education-focused applications were uncommon (n=22, 7.5%)(UpSet Plot).

Machine learning was the most frequent AI method (n=202, 69.2%), followed by deep learning (n=75, 25.7%), computer vision or image analysis (n=51, 17.5%), and neural networks (n=40, 13.7%). The most frequent conditions or topics were sepsis, COVID-19, tuberculosis, respiratory infection, bloodstream infection, antimicrobial resistance, and organism identification (Heatmap). Educational applications chiefly employed commercially available Large Language Models (13/22) (Supplementary Material)

DISCUSSION / CONCLUSION

AI research in infectious diseases and clinical microbiology is expanding rapidly, with most included studies published from 2020 onwards. Although the evidence base is dominated by diagnostic machine-learning studies, particularly in sepsis, the range of applications is broad, spanning clinical prediction, imaging, microscopy, antimicrobial resistance, organism identification, antimicrobial stewardship, public health, large language models, and education. Several studies also made code or analytical workflows available, highlighting opportunities for reproducibility, external validation, and local adaptation. The review shows that AI in infectious diseases and microbiology is best understood not as a single technology, but as a diverse methodological toolkit being applied unevenly across a rapidly expanding evidence base. The next challenge is not simply to build more models, but to develop transparent, validated, clinically useful AI tools that address real-world priorities in infection diagnosis, treatment decision-support, microbiology workflows, and education.

REFERENCES AND ACKNOWLEDGEMENTS

References:

1. Topol EJ. High-performance medicine: the convergence of human and artificial intelligence. Nat Med. 2019;25(1):44-56. doi:10.1038/s41591-018-0300-7.
2. Yu KH, Beam AL, Kohane IS. Artificial intelligence in healthcare. Nat Biomed Eng. 2018;2(10):719-731. doi:10.1038/s41551-018-0305-z.
3. Wong F, de la Fuente-Nunez C, Collins JJ. Leveraging artificial intelligence in the fight against infectious diseases. Science. 2023;381(6654):164-170. doi:10.1126/science.adh1114.
4. Peiffer-Smadja N, Dellière S, Rodriguez C, Birgand G, Lescure FX, Fourati S, et al. Machine learning in the clinical microbiology laboratory: has the time come for routine practice? Clin Microbiol Infect. 2020;26(10):1300-1309. doi:10.1016/j.cmi.2020.02.006.
5. Abbara S, Crabol Y, Goupil de Bouillé J, Dinh A, Morquin D, Groupe Infectiologie Digitale-SPILF. Artificial intelligence and infectious diseases: scope and perspectives. Infect Dis Now. 2025;55(7):105131. doi:10.1016/j.idnow.2025.105131.
6. Peters MDJ, Marnie C, Tricco AC, Pollock D, Munn Z, Alexander L, et al. Updated methodological guidance for the conduct of scoping reviews. JBI Evid Implement. 2021;19(1):3-10. doi:10.1097/XEB.0000000000000277.
7. Ouzzani M, Hammady H, Fedorowicz Z, Elmagarmid A. Rayyan-a web and mobile app for systematic reviews. Syst Rev. 2016;5:210. doi:10.1186/s13643-016-0384-4.

The researchers would like to acknowledge the invaluable support of Shona Nolan, Librarian at University Hospital Waterford, who assisted in refining the search strategy and translating the MEDLINE search approach for use in Web of Science and Embase.